

Design and Implementation of a Secure RFID-Based Access Control System

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—Abstract— Schools continue to explore new ways to improve security and attendance tracking systems. Traditional attendance methods are vulnerable to human error and inaccurate attendance records. The project focuses on developing an RFID-based attendance management system using a Raspberry Pi, an RFID scanner, and a web-based platform. The objective of the project is to improve the organization of attendance, accuracy, and security within the Brooklyn STEAM Center.

Keywords — Radio Frequency Identification(RFID), Raspberry Pi, Attendance Management, Access Control, Cybersecurity

I. INTRODUCTION

As schools continue to grow and manage larger numbers of students, maintaining accurate attendance records and security becomes difficult, creating attendance gaps when security is treated as an afterthought. The Brooklyn STEAM center currently lacks an automated attendance-logging system. This likely leaves gaps between student attendance and security's ability to identify students. To combat this risk, RFID (Radio-Frequency Identification) has been proposed. RFID is a wireless technology that uses radio waves to identify and track objects. At the same time, Near Field Communication (NFC), a subset of RFID, enables short-range communication with devices via a chip in a card or badge. The time a student walks in can never be fabricated because attendance is collected automatically via a badge and identity. In the work environment, RFID technology can assist in investigations involving both insider and outsider threats. This makes it a more reliable way to track attendance and help schools and workplaces keep accurate records while improving overall security. Previous attempts to implement an automated RFID system with similar resources were unsuccessful due to hardware limitations and software compatibility issues, which created roadblocks for the group.

II. LITERATURE REVIEW

A. Problem / Background Research

Across schools and organizations, manual attendance systems such as paper registrations and weak access controls can lead to

inaccurate records, fraud, and even serious security breaches, highlighting the need for more reliable logging and badge-based solutions. Studies on attendance systems indicate that manual methods are time-consuming and error-prone, while newer technologies, such as automated tracking and facial recognition, aim to improve efficiency and accuracy [1]. In real-world situations, weak logging and access control have already caused major issues, such as the Target data breach, where attackers gained access through a third-party vendor without proper monitoring [2]. Similarly, investigations into school systems, such as the Chicago Public Schools attendance fraud report, found that attendance records were being manipulated, proving how unreliable manual systems can be [3]. These sources show that current systems lack accuracy, security, and reliability.

B. Project Focus

In response to these issues, research shows that automated attendance and logging systems provide a more efficient and safer alternative. Technologies such as RFID, facial recognition, and digital tracking systems enable automatic attendance recording, reducing human error and preventing proxy attendance [1]. These systems also create real-time logs that can be used to monitor individuals, improve accountability, and support investigations when needed. In various workplace settings, tracking attendance can increase productivity and responsibility among students and employees, as it signals that their presence is being monitored [4].

III. PROJECT REQUIREMENTS

The following materials were used in the development of this project:

- RFID Reader
- NFC Cards (50x)
- Display screen
- Raspberry Pi+++
- SD Card (16GB)

The materials used for the experiment are as follows: RFID Scanner (16 GB SD card), Raspberry Pi, and Display Screen.

IV. METHODOLOGY

The team will use VS Code to write and test Python scripts for the Raspberry Pi by connecting the Raspberry Pi to an RFID scanner (serial). The data will be stored and processed when a card is scanned. Additional scripts will allow the team to collect other types of information.

Storing information is extremely important to the process. Google Sheets will be used to store students' entry times and mark them as late, on time, absent, or present, with an additional "excused" option entered manually. A CSV database will be used to automate the scanning process to Google Spreadsheets. This will be stored on the 16-gigabyte SD card. Due to the nature of this experiment, no real student information will be used during testing. Thus, all tested information will be simulated, including fake names, fake student IDs, and fake photos. This ensures that the system will be capable of handling real student information and will be implemented in future school years.

VI. EXPERIMENT

A. Initial Hardware Testing During the experimentation process, the Raspberry Pi and RFID scanner were configured and tested multiple times to verify their functionality. RFID cards were connected and tested to confirm that the hardware functioned correctly. RFID cards were scanned multiple times to verify that the system could successfully detect and display student IDs.

B. Attendance System Development After the team confirmed that the RFID scanner functions correctly. Attendance records, timestamps, and student information were integrated into the attendance system. Python scripts and RFID libraries were used to process attendance data and organize student information.

C. Troubleshooting & Technical Issues During the development phase, several technical issues involving the Python libraries, JSON configuration files, scanner detection, and most importantly, Google Sheets integration were encountered. Despite the RFID card detection functioning correctly, the system experienced communication and credential errors when attempting to connect the attendance records to Google Sheets.

D. System Redesign Due to the continuous technical issues involving Google Sheets, the attendance system was redesigned into a website-based platform. The redesign system introduced separate account roles for teachers, admins, and super admins, while also adding live attendance monitoring, attendance records, attendance statistics, and student management features

E. Final Testing After redesigning the system, RFID scans were successfully shown through the website in real time: attendance records, timestamps, student information, attendance statistics, and student management features.

VII. RESULTS

The redesigned attendance system successfully detected RFID card scans and recorded student attendance information in real time. During testing, multiple RFID cards could be scanned, and the website continuously displayed student IDs, timestamps, and attendance status. The system also successfully installed separate teacher, admin, and super admin roles, which allow attendance records to be managed by different user types. Testing showed that attendance information appears immediately after a scan is completed and that the system is more organized than manual attendance tracking. However, there were also limitations, such as limited long-term testing and the need for improvements in scanner speed and system reliability.

VIII. CONCLUSION & FUTURE WORKS

Developing and implementing the RFID attendance system for students and staff was a difficult process that required the team to work together and solve problems throughout the experiment. During the project, the team successfully created a website-based attendance management system connected to the RFID scanner and Raspberry Pi.

The system still has areas for exploration and improvement, such as implementing MFA for staff accounts and converting the browser to HTTPS; the experiment allowed the team to gain experience with RFID technology, website development, and database management. Future development may include mobile device integration, cloud-based data storage, stronger authentication control, faster scanner response times, and support for classes and users within a school environment.

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